Nested GLS Profile Spectrum: First Smoke Test

June 21, 2026

Purpose

This note records what is needed to compute the full regular eigenspectrum for nested tests in the profile-Hessian paradigm, and a first finite-sample smoke test in a continuous GLS setting. The companion scripts are

`relative-and-nested-fit-check.py` and `gls_nested_profile_mc.py`.

1 Required Objects

Let x be the first-stage vector and let

$$\sqrt{n}(\hat{x} - x_0) \Rightarrow Z, \qquad Z \sim N(0, \Gamma_x).$$

For each model M, define the profiled value

$$v_M(x) = \min_\theta f_M(\theta, x).$$

The nested contrast is

$$c(x) = v_B(x) - v_A(x),$$

where B is the restricted model and A is the larger model. Under the regular nested pseudo-null,

$$c(x_0) = 0, \qquad \nabla c(x_0) = 0.$$

The reference law is therefore

$$n\, c(\hat{x}) \Rightarrow \frac{1}{2} Z'(Q_B - Q_A) Z$$

if f is scaled as a half-discrepancy, or equivalently

$$2n\, c(\hat{x}) \Rightarrow Z'(Q_B - Q_A) Z.$$

Thus the full spectrum is obtained from the self-adjoint version of

$$(Q_B - Q_A)\Gamma_x.$$

In implementation terms, the steps are:

1. Build a common first-stage vector x and covariance Γ_x.
2. Compute the profiled score $s_M(x) = \nabla_x v_M(x)$ for each model.
3. Differentiate it to get $Q_M = D_x s_M(x_0)$.
4. Verify the pseudo-null conditions $v_B = v_A$ and $s_B = s_A$.
5. Eigensolve $\Gamma_x^{1/2}(Q_B - Q_A)\Gamma_x^{1/2}$.

The old $U_B - U_A$ spectrum is the special exact-fit/fixed-weight/Gauss–Newton case. It is not the general object.

2 GLS Smoke Design

The smoke uses the linear covariance example from `relative-and-nested-fit.tex`. There are four observed variables.

$$B: \text{ compound symmetry: one variance and one covariance,}$$

$$A: \text{ one common variance and free covariances.}$$

The test is equality of the covariances. The normal-theory GLS weight is estimated from the first-stage covariance:

$$W(u) = \Gamma_{\mathrm{NT}}(u)^{-1}.$$

To make the nested pseudo-null exact while keeping both models misspecified, the population covariance is diagonal with unequal variances:

$$\Sigma_0 = I + \epsilon\,\mathrm{diag}(d), \qquad d = (1, -1, 1, -1).$$

Both models share the same pseudo-true zero covariance structure, so

$$v_B(x_0) - v_A(x_0) = 0, \qquad s_B(x_0) - s_A(x_0) = 0,$$

but the common-variance part is misspecified when $\epsilon \neq 0$.

3 Population Spectrum

For $\epsilon = 0.15$, the population script reports:

$$v_B - v_A = 0, \qquad \|s_B - s_A\| = 0,$$

and the profile/value finite-difference Hessians agree to 1.17×10^{-7}. The classical fixed-weight spectrum is

$$(1, 1, 1, 1, 1),$$

whereas the profile-Hessian spectrum is

$$(1.4082,\ 0.9139,\ 0.9139,\ 0.9139,\ 0.5057), \qquad \mathrm{tr} = 4.6557.$$

So even in this linear model with no model-curvature term, estimating the GLS weight changes the reference law under fixed misspecification.

4 Finite-Sample Smoke

The Monte Carlo uses $N = 1200$ and 6000 replications, simulating normal data from the same Σ_0 and refitting both GLS profiles with $W(\hat{u})$.

Reference	q50	q90	q95	q99
Empirical statistic	3.9382	8.8421	10.7279	15.4107
Profile spectrum	3.9829	8.7284	10.5929	14.7434
Classical U / χ^2_5	4.3539	9.2525	11.0753	14.9859

The empirical tail probability at the profile-spectrum 95% cutoff was 0.0532. At the ordinary χ^2_5 cutoff it was 0.0450. This is only a smoke test, but it points in the expected direction: the profile-Hessian spectrum tracks the finite-sample statistic better than the exact-fit spectrum.

5 What This Suggests

For GLS, the full nested eigenspectrum is not hard in principle. The hard part is not the eigensolve; it is computing Q_M reliably. Finite differences of the profile score are enough for research scripts. A production implementation should expose analytic or automatic profile derivatives for:

$$s_M(x), \qquad Q_M = D_x s_M(x).$$

For categorical DWLS, the same steps apply, but x must include the weight inputs such as $\gamma = \mathrm{diag}(\Gamma_u)$. That makes Q_M a block Hessian over (u, γ) rather than only over ordinal moments. The GLS smoke is therefore the right first implementation target: it validates the profile machinery before the ordinal first-stage stack is made larger.